

VISHAL R B

Blockchain Developer | Web3 Full-Stack Developer | AI & ML Engineer | Frontend Developer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Versatile Web3 Full-Stack Developer, Blockchain Developer, and AI & ML Engineer specializing in building secure DApps, intelligent predictive analytics, and high-performance user interfaces. Highly proficient as a Frontend Developer utilizing Next.js 15/16, React.js, and Tailwind CSS, backed by strong backend and smart contract architecture using Solidity, Hardhat, IPFS, Django, and custom machine learning pipelines. Experienced in Web wallet security, multi-sig asset management, and end-to-end decentralized application engineering.

TECHNICAL SKILLS

Blockchain & Web3: Solidity, Ethereum, Hardhat, Smart Contracts, Viem, Chainlink, Pinata (IPFS).

AI & Machine Learning: Gemini API, XGBoost, ChromaDB, Vector Embeddings, Predictive Analytics, NLP Pipelines.

Computer Vision: OpenCV, Real-Time Video Processing, Object Detection, Image Processing.

AI Tools & Platforms: ChatGPT, Claude Code, Google Gemini, Perplexity AI, Antigravity, Codex.

Frontend Development: Next.js 15/16 (App Router), React.js 19, TypeScript, JavaScript (ES6+), Tailwind CSS, HTML5, CSS3.

Backend Development: Node.js, Express, Python, Django, Django REST Framework, Celery, Redis, RESTful APIs.

Databases & Storage: PostgreSQL, SQLite, MongoDB.

Tools & Environments: Git, GitHub, Gitpod, Vercel, Render, Kali Linux Terminal, Postman.

EXPERIENCE

QBrainly

Full-Stack Development — AI Education Platform

Collaborative Project

Remote

- Contributed to a monorepo full-stack platform featuring a Django REST backend and Next.js 16 App Router frontend, supporting role-based access for Superadmins, Faculty, and Students.
- Connected Next.js UI components to DRF endpoints using unified API utilities, enforcing secure request handling with CSRF protection and session authentication.
- Designed Django serializers and views across role-specific namespaces (`/api/students`, `/api/faculty`, `/api/superadmin`) to ensure strict, clean JSON contract alignment.
- Collaborated on asynchronous processing pipelines using Celery and Redis alongside AI-enabled content extraction, chunking, and ChromaDB vector embedding flows.

Ramco Cements

Machine Learning Intern — Mill Residue Prediction System

3 Months

Rajapalayam, India

- Developed an end-to-end machine learning pipeline using XGBoost to forecast industrial mill residue levels from operational parameters, reaching ~ 91% prediction accuracy on validation sets.
- Preprocessed, transformed, and cleaned over 10,000 raw industrial sensor and process data records for machine learning model ingestion.
- Built predictive analytics workflows that enhanced early detection of residue deviations, reducing manual process monitoring effort by ~ 30%.
- Generated automated predictive control suggestions from model outputs to provide real-time decision support for industrial operations.

Question Paper Generator (QPG)

Frontend Development — AI Question Paper Generation

3 Months

Remote

- Built responsive web application interfaces and layouts using Next.js 16, React 19, and Tailwind CSS, designing secure token-based session routing to streamline startup UX.
- Connected the frontend to a Django REST API using Axios, implementing session authentication, CSRF security, and model serialization for exam patterns and generated tests.
- Designed interactive feature interfaces and admin views using modular React components and Lucide icons following maintainable project architecture patterns.

- Integrated client-side **docx-preview** capabilities to allow users to inspect rendered question paper files directly in the browser prior to exporting.

PROJECTS

AI-Powered Crypto Portfolio Tracker

AI & ML / Web3 Full-Stack Project / Ramco Institute of Technology

- Engineered a full-stack Web3 application using Next.js and TypeScript, leveraging Server-Side Rendering (SSR) to ensure real-time multi-wallet asset data accuracy.
- Integrated AI & ML capabilities via the Gemini API to deliver automated market sentiment analyses and automated portfolio rebalancing strategies.
- Designed responsive, data-dense visualizations in Tailwind CSS, boosting user data readability and portfolio tracking efficiency by 40%.

Decentralized Voting System 2.0

Blockchain & Frontend Development / Ramco Institute of Technology

- Architected a Glassmorphism-inspired interface using Next.js 15 and Tailwind CSS, creating a seamless onboarding experience for non-technical users.
- Integrated frontend components with Viem and Solidity smart contracts, cutting transaction latency and enhancing wallet connection resilience.
- Implemented Pinata IPFS metadata storage to maintain decentralized, immutable candidate profiles and election records.

Decentralized NFT Marketplace (MetaMarket)

Blockchain & Smart Contract Project / Ramco Institute of Technology

- Developed and deployed custom ERC-721 smart contracts in Solidity using Hardhat for local compiling, unit testing, and testnet deployment.
- Managed decentralized storage pipelines using Pinata (IPFS) for immutable digital artwork metadata hosting.
- Built an interactive React frontend optimizing dynamic array rendering to process secondary market listings smoothly.

EDUCATION

Ramco Institute of Technology

B.Tech in Artificial Intelligence & Data Science

2023 – 2027

Rajapalayam, Tamil Nadu

S.N.M. Hindu Vidyalaya Matriculation Higher Secondary School

Secondary & Higher Secondary Education

2016 – 2022

Nagercoil, Tamil Nadu

CERTIFICATIONS

- **Certificate Program in Artificial Intelligence & Machine Learning – Wipro**
- **Certificate Program in Cloud Computing Engineering – Wipro**
- **Blockchain Basics Course – Cyfrin Updraft**
- **JavaScript Course With Certification – Scalar School**

ACHIEVEMENTS

- Awarded **1st Place** at KPRIET for engineering an innovative project combining AI with scalable Distributed Computing.
- Recognized by Rise In for completing the intensive **Stellar Journey to Mastery Program**.
- Earned the **Advanced Web3 Wallet Security Badge** from Cyfrin Updraft for mastering smart contract and asset defense.